Beyond scalar inertia

Specialised extension: general 3-D rotation. The JEE sequence needs only planar and fixed-axis components.

Tensor definition

$$I_{ij} = \int (r^2 \delta_{ij} - x_i x_j)\, dm$$

$$\vec{L}_C = \mathbf{I}_C \vec{\omega}$$

$$K_{\text{spin}} = \tfrac{1}{2} \vec{\omega} \cdot \mathbf{I}_C \vec{\omega}$$

Coordinates are relative to C; delta(ij) is 1 for equal indices, 0 otherwise. I is symmetric and has orthogonal principal axes.

WHY / Each mass contributes r cross (omega cross r). Collecting coefficients gives this linear map.

Dumbbell: a clean counterexample

Two point masses m at ±a n, where n is a unit vector along a massless rod.

$$\mathbf{I} = 2ma^2(\mathbf{1} - \vec{n}\vec{n}^{\,T})$$

$$\vec{L} = 2ma^2[\vec{\omega} - (\vec{\omega} \cdot \vec{n})\vec{n}]$$

WHY / A rotation component along the zero-thickness rod contributes no angular momentum. L is perpendicular to n.

WATCH / OL8's component expression uses a non-orthogonal direction incorrectly. For generic oblique spin, L is not parallel to omega.

Body frame versus lab frame

$$\vec{\tau}_C = \left(\frac{d\vec{L}_C}{dt}\right)_{\text{body}} + \vec{\omega} \times \vec{L}_C$$

In principal axes the body-frame inertia matrix is diagonal. The extra cross term accounts for rotation of the basis itself.

WHY / A vector fixed in the body can still change direction in the laboratory.

WATCH / Do not differentiate body-frame components as if their basis vectors were fixed in space.

An arbitrary moving origin

$$\vec{L}_A = \sum_i (\vec{r}_i - \vec{r}_A) \times m_i \vec{v}_i$$

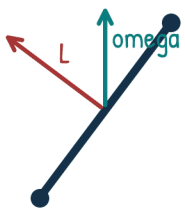
$$\frac{d\vec{L}_A}{dt} = \vec{\tau}_A - \vec{v}_A \times M\vec{v}_C$$

These velocities are absolute inertial velocities. For A=C the cross term vanishes. For relative velocities, the corresponding rate formula instead involves a(A).

WHY / Differentiating r(i)-r(A) generates the transport term.

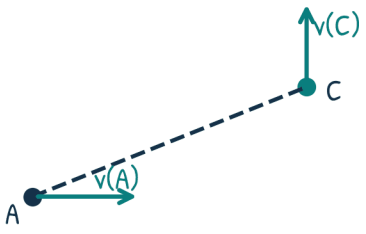
WATCH / A moving axis is not automatically a conserved-L origin just because its external torque vanishes.

Oblique dumbbell: subtract the component along the rod



L dot n = 0

Moving origins require the angular-momentum transport term



REMEMBER / The scalar I about an axis remains useful even when the full angular-momentum vector is not parallel to omega.

Gyroscopes: torque turns L

Specialised symmetric-top dynamics; separate spin, precession and nutation.

Geometry removes sin(theta)

Fixed pivot O; C at distance d along symmetry axis n. L is approximately I3 omega3 n for dominant spin.

$$|\tau| = Mgd\sin\theta$$

$$|\dot{\vec{L}}| = \Omega L\sin\theta$$

$$\Omega \simeq \frac{Mgd}{I_3\omega_3}$$

WHY / Both torque and the precession-circle radius contain sin(theta). They cancel at nonzero tilt.

WATCH / Use tau/(L sin(theta)), not tau/L for the precession rate around vertical.

When slow precession is safe

A nearly symmetric top, spin angular momentum dominant, slow steady precession and small nutation amplitude. Finite tilt is allowed.

$$I_\perp \Omega \ll |I_3\omega_3|$$

I(perp) is the transverse inertia about the pivot, not the COM. Initial conditions can excite nutation even at modest tilt.

WHY / Rapid spin gives a large L, so a given torque turns it slowly.

WATCH / As spin tends to zero the approximation fails; the predicted divergence does not mean an instantaneous fall.

Steady-precession branches

$$I_\perp \Omega^2\cos\theta - I_3\omega_3\Omega + Mgd = 0$$

For a heavy symmetric top in regular precession, theta is constant; omega3 is the actual angular velocity component along n.

WHY / The quadratic retains transverse angular momentum. The small root reduces to slow precession when spin dominates.

WATCH / Reality of roots and initial conditions matter. This is not the same as stability of an upright sleeping top.

Upright "sleeping" top

Ideal heavy symmetric top, fixed point, C above O, upright rotation; small tilt perturbations.

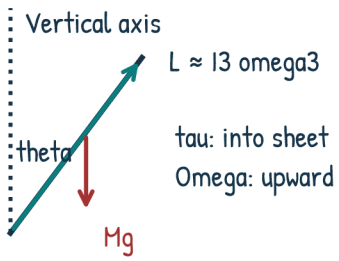
$$(I_3\omega_3)^2 > 4Mgd\,I_\perp$$

WHY / Linearised tilt equations give bounded oscillatory perturbations above this threshold; equality is marginal.

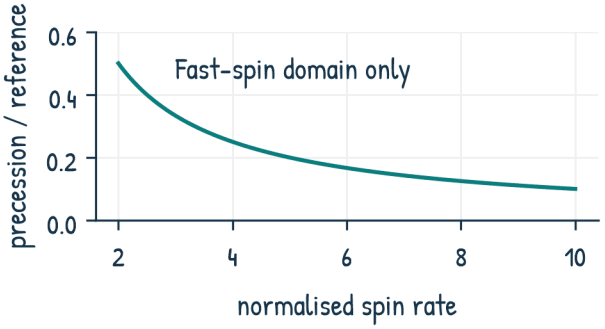
WATCH / No universal extra R/r factor exists. Real tip friction, damping and geometry require their own model.

Check E6: Omega≈0.25 rad/s; precession period≈25.13 s. Fast spin alone does not specify initial nutation.

Gravity turns spin angular momentum around vertical



Inverse-spin trend: do not extrapolate to zero spin



REMEMBER / Slow precession is a fast-spin approximation, not a small-tilt law; omega->0 is outside that approximation.

Flywheel limits: add a material law

Kinetic-energy storage is JEE core; elastic stress and failure models are specialised extensions.

Core energy, new restriction

$$K = \tfrac{1}{2} I\omega^2$$

$$\frac{K}{M} = \tfrac{1}{4}(\omega R)^2 \quad \text{(solid disc)}$$

For a prescribed spin rate this needs only rigid-body mechanics. A maximum allowable speed needs stress information.

WHY / Scaling to energy per mass removes size, but does not eliminate the need for a material strength limit.

WATCH / Do not call all flywheel energy questions "outside JEE"; it is the material theory that is additional.

Uniform elastic thin disc

Ideal isotropic constant-thickness disc, free outer edge, no centre hole, plane-stress linear elasticity.

$$\sigma_{\max} = \frac{3+\nu}{8}\rho\omega^2 R^2$$

$$\left(\frac{K}{M}\right)_{\max} = \frac{2\sigma_{\text{allow}}}{\rho(3+\nu)}$$

rho is density; nu is Poisson's ratio. Maximum stress is at the centre for ordinary positive nu.

WHY / Insert the stress-limited rim speed into K/M. Dimensions: stress/density = m²/s².

Numerical scale

rho=7800 kg/m³, nu=0.30, sigma(allow)=500 MPa; use the source's ideal strength as the allowable value.

$$(\omega R)_{\max} = 394.2 \text{ m/s}$$

$$K/M = 38.85 \text{ kJ/kg}$$

WHY / This independently recomputes the source's approximately 39 kJ/kg estimate.

WATCH / An engineering allowable stress usually includes safety factors, fatigue and manufacturing defects. This is not a safe design recommendation.

Changing shape changes stress

For an ideal thin rotating ring the hoop stress is rho omega² R², while K/M = omega² R² / 2.

$$(K/M)_{\max,\,ring} = \sigma_{\text{allow}}/(2\rho)$$

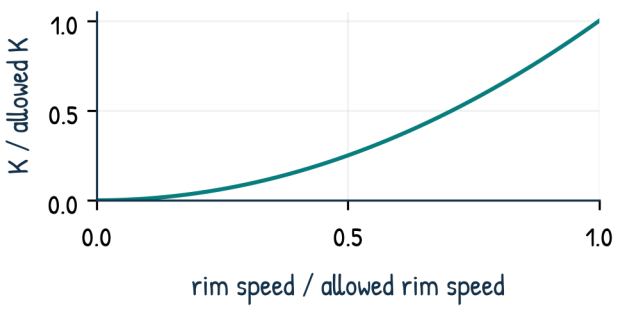
WHY / The ring stores more at fixed rim speed but reaches its own stress limit sooner; compare under the same failure condition.

WATCH / A bored disc, composite rotor, anisotropy, bearing mass or enclosure mass changes the estimate. A battery comparison is not a universal performance ranking.

Elastic disc model: no bore, isotropic, plane stress



Both kinetic energy and elastic stress scale as spin squared



REMEMBER / A perfectly rigid model cannot itself predict breaking speed: a constitutive and failure model must be supplied.

Open systems & missing constraints

Specialised model audit: not every printed question is sufficiently specified to have a unique answer.

Unrolling carpet: variable mass

A real roll has finite thickness t; a thin-layer continuum may approximate its cross-section. Negligible thickness literally cannot create a finite-radius roll.

$$m(R) \simeq MR^2/R_0^2$$

$$I_C(R) \simeq \tfrac{1}{2}m(R)R^2$$

WHY / With a negligible core and uniform packing, remaining mass scales with cross-sectional area.

WATCH / The rolled part is an open subsystem: momentum and energy flow out when layers are deposited.

What a complete model needs

Specify initial packing, core radius, thickness, release mechanism, no-slip contacts, and whether deposited material is stopped inelastically.

$$K_{\text{roll}} = \tfrac{3}{4}m(R)v_C^2$$

This expression assumes a uniform cylindrical roll and omega=v(C)/R. It is only an instantaneous kinetic energy, not an energy balance.

WHY / The whole-carpet energy must include changing potential, kinetic energy of every part, and any deposition loss.

WATCH / OL5 does not specify these laws. No unique speed-versus-radius result is asserted here.

Rod and peg: clarify the stages

A smooth floor supplies only vertical force; it cannot generally pin the rod's bottom. A peg contact is not automatically a permanent hinge.

$$\vec{L}_P^{\,-} \simeq \vec{L}_P^{\,+}$$

Across a short single-point peg impact this holds if other external angular impulses about fixed P are negligible.

WHY / Impulse at P has zero moment about P, but another simultaneous support impulse need not.

WATCH / OL9 lacks peg location/contact geometry, capture and restitution rules. Its assumed fixed-bottom fall contradicts a smooth floor.

After repairing the model

If a fixed pin captures the body at P, calculate pre-impact v(C) and omega, then L(P), then post-impact KE with the new inertia.

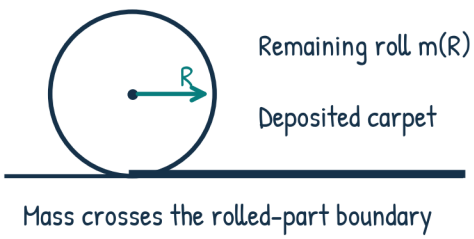
$$K_+ = \frac{L_P^2}{2I_P}$$

Only afterwards compare with the gravitational barrier. For unilateral contact, also test whether the peg reaction remains admissible.

WHY / This is the same phase logic used for the hinged bullet problem, but with new constraints explicitly supplied.

WATCH / No numerical swing-over condition can be certified from the source OL9 statement alone.

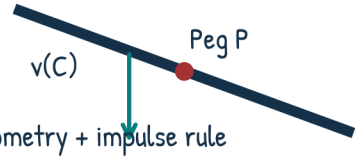
Open rolling subsystem: deposition needs a physical law



Mass crosses the rolled-part boundary

Peg impact is not automatically a new permanent hinge

Is contact captured or unilateral?



Geometry + impulse rule
+ later contact test

REMEMBER / If the model lacks a deposition law, contact geometry or impact rule, state that rather than inventing a final formula.